

HDOC – 22nd day

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Aim:

To make a menu driven C program that will take two integers as input two integers and display their multiplication table, a multiplication grid and consecutive integers until the user chooses the exit option.

PROBLEM STATEMENT:

Write a menu-driven C program that accepts two positive integers, m and n, where $m \leq n$. Display the following menu and use a switch statement to perform the operation selected by the user:

1. Print the multiplication tables of m and n side by side, using multipliers from 1 to 10.
2. Display a multiplication grid containing the products of integers from 1 to m and integers from 1 to n.
3. For each integer from m to n, print that integer followed by the next five consecutive integers.
4. Exit the program.

❖ Algorithm:

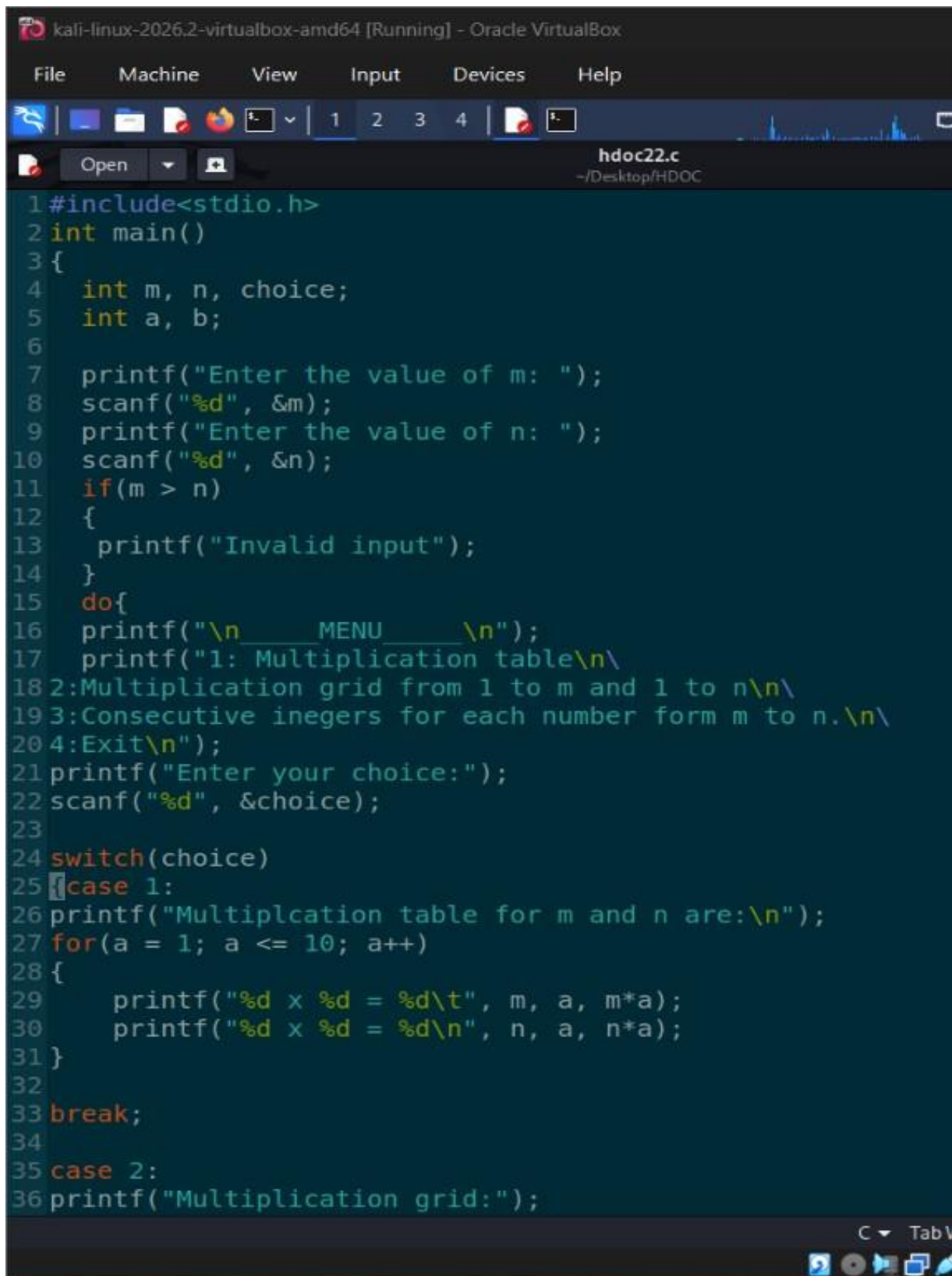
- 1) Start.
- 2) Declare m, n, a, b, choice as integer.
- 3) Take input m and n.
- 4) If input is invalid(m is not less than equal to n) then terminate the program.
- 5) Display a menu of choices:
 - a) Multiplication table.

- b) multiplication grid from 1 to m and 1 to n.
 - c) consecutive integers for each number from m to n.
 - d) Exit
- 6) Use a switch statement.
- 6.1) For Choice 1, use a loop from 1 to 10 and print the multiplication tables of m and n side by side.
- 6.2) For Choice 2, use nested loops to calculate and display the products of integers from 1 to m with integers from 1 to n.
- 6.3) For Choice 3, use nested loops. For every integer from m to n, print that integer followed by its next five consecutive integers.
- 6.4) For Choice 4, terminate the program.
- 7) For any other choice, display "Invalid Choice".
- 8) Repeat the menu until the user selects Choice 4 (Exit).
- 9) Stop the program.

Test Case Table:

Test Case	Input	Choice	Expected output	Actual output	Result
1.	M = 2 N = 3	1	Multiplication table of 2 and 3	Multiplication table of 2 and 3	Pass
2.	M = 1 N = 4	2	Multiplication grid of 1 to 1 and 1 to 4	Multiplication grid of 1 to 1 and 1 to 4	Pass
3.	M = 3 N = 4	3	4, 5, 6, 7, 8	4, 5, 6, 7, 8	Pass
4.	M = 1 N = 10	4	Exiting the program	Exiting the program	Pass
5.	M = 2 N = 8	5	Invalid choice	Invalid choice	Pass

C Program:



The image shows a terminal window titled "kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox". The window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". Below the menu bar is a toolbar with icons for file operations. The terminal displays the source code for a C program named "hdoc22.c" located at "~/Desktop/HDOC". The code is as follows:

```
1#include<stdio.h>
2int main()
3{
4    int m, n, choice;
5    int a, b;
6
7    printf("Enter the value of m: ");
8    scanf("%d", &m);
9    printf("Enter the value of n: ");
10   scanf("%d", &n);
11   if(m > n)
12   {
13       printf("Invalid input");
14   }
15   do{
16       printf("\n_____MENU_____\n");
17       printf("1: Multiplication table\n\
182:Multiplication grid from 1 to m and 1 to n\n\
193:Consecutive inegers for each number form m to n.\n\
204:Exit\n");
21printf("Enter your choice:");
22scanf("%d", &choice);
23
24switch(choice)
25{case 1:
26printf("Multiplcation table for m and n are:\n");
27for(a = 1; a <= 10; a++)
28{
29    printf("%d x %d = %d\t", m, a, m*a);
30    printf("%d x %d = %d\n", n, a, n*a);
31}
32
33break;
34
35case 2:
36printf("Multiplication grid:");
```

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

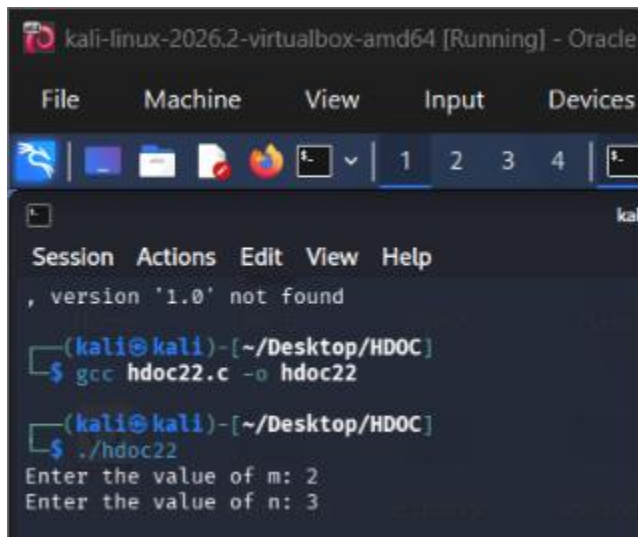
File Machine View Input Devices Help

hdoc22.c
~/Desktop/HDOC

```
34
35 case 2:
36 printf("Multiplication grid:");
37
38 {for(a = 1; a<=m; a++)
39 {
40     for(b = 1; b<=n; b++)
41     printf("%d x %d = %d\n", a, b, a*b);
42 }
43
44 }
45 break;
46 case 3:
47 printf("Consecutive integer:");
48 {for(a = m; a <= n; a++)
49 {
50     for(b = a; b <= a+5; b++)
51     {
52     printf("%d\n", b);
53     }
54 }
55 break;
56 }
57 case 4:
58 printf("Exiting the program");
59 break;
60
61 default:
62 printf("Invalid choice\n");
63 break;
64 }
65 }
66 }while(choice != 4);
67 return 0;
68
69 }
```

Compile the code using:

```
gcc hdoc22.c -o hdoc22
```



The screenshot shows a terminal window titled "kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle". The terminal has a menu bar with "File", "Machine", "View", "Input", and "Devices". Below the menu bar is a toolbar with icons for file operations and a tab bar with tabs labeled "1", "2", "3", "4", and a terminal icon. The terminal content shows the following commands and output:

```
Session Actions Edit View Help
, version '1.0' not found

(kali@kali)-[~/Desktop/HDOC]
$ gcc hdoc22.c -o hdoc22

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc22
Enter the value of m: 2
Enter the value of n: 3
```

Execute the code using:

```
./hdoc22
```

Here are some screenshots of execution of code against test case

```
(kali@kali)-[~/Desktop/HDOC]
$ gcc hdoc22.c -o hdoc22

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc22
Enter the value of m: 2
Enter the value of n: 3

____MENU____
1: Multiplication table
2: Multiplication grid from 1 to m and 1 to n
3: Consecutive integers for each number from m to n.
4: Exit
Enter your choice: 1
Multiplication table for m and n are:
2 x 1 = 2      3 x 1 = 3
2 x 2 = 4      3 x 2 = 6
2 x 3 = 6      3 x 3 = 9
2 x 4 = 8      3 x 4 = 12
2 x 5 = 10     3 x 5 = 15
2 x 6 = 12     3 x 6 = 18
2 x 7 = 14     3 x 7 = 21
2 x 8 = 16     3 x 8 = 24
2 x 9 = 18     3 x 9 = 27
2 x 10 = 20    3 x 10 = 30

____MENU____
1: Multiplication table
2: Multiplication grid from 1 to m and 1 to n
3: Consecutive integers for each number from m to n.
4: Exit
Enter your choice: 4
Exiting the program
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~/D
Session Actions Edit View Help
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc22
Enter the value of m: 1
Enter the value of n: 4

MENU
1: Multiplication table
2: Multiplication grid from 1 to m and 1 to n
3: Consecutive integers for each number from m to n.
4: Exit
Enter your choice: 2
Multiplication grid: 1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
1 x 4 = 4

MENU
1: Multiplication table
2: Multiplication grid from 1 to m and 1 to n
3: Consecutive integers for each number from m to n.
4: Exit
Enter your choice: 4
Exiting the program

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc22
Enter the value of m: 3
Enter the value of n: 4

MENU
1: Multiplication table
2: Multiplication grid from 1 to m and 1 to n
3: Consecutive integers for each number from m to n.
4: Exit
Enter your choice: 3
Consecutive integer: 3
4
5
6
7
8
Exiting the program

MENU
1: Multiplication table
2: Multiplication grid from 1 to m and 1 to n
3: Consecutive integers for each number from m to n.
4: Exit
Enter your choice: 4
Exiting the program
```

```
(kali@kali)-[~/Desktop/HDOC]
```

```
$ ./hdoc22
```

```
Enter the value of m: 1
```

```
Enter the value of n: 10
```

```
____MENU____
```

```
1: Multiplication table
```

```
2: Multiplication grid from 1 to m and 1 to n
```

```
3: Consecutive integers for each number from m to n.
```

```
4: Exit
```

```
Enter your choice: 4
```

```
Exiting the program
```

```
(kali@kali)-[~/Desktop/HDOC]
```

```
$ ./hdoc22
```

```
Enter the value of m: 2
```

```
Enter the value of n: 8
```

```
____MENU____
```

```
1: Multiplication table
```

```
2: Multiplication grid from 1 to m and 1 to n
```

```
3: Consecutive integers for each number from m to n.
```

```
4: Exit
```

```
Enter your choice: 5
```

```
Invalid choice
```

```
____MENU____
```

```
1: Multiplication table
```

```
2: Multiplication grid from 1 to m and 1 to n
```

```
3: Consecutive integers for each number from m to n.
```

```
4: Exit
```

```
Enter your choice: 4
```

```
Exiting the program
```

```
(kali@kali)-[~/Desktop/HDOC]
```

```
$
```